Function operations

1 Given that $f(x) = x^2 - 1$ and $g(x) = 5x - 1$, find $f(2) + g(2)$.

2 Consider $f(x) = x^2 + 6$ and $g(x) = 5x - 3$.

a Find $(f - g)(x)$.

b Evaluate $(f - g)(5)$.

3 Consider $f(x) = -5x + 3$ and $g(x) = x^2 - 7$.

a Find $(f \cdot g)(x)$.

b Evaluate $(f \cdot g)(-3)$.

4 Given that $f(x) = 3x - 5$ and $g(x) = 5x + 7$, find the following:

a $(f + g)(x)$

b $(f + g)(4)$

c $(f - g)(x)$

d $(f - g)(10)$

5 Given the table of values, find $(f \cdot g)(6)$.

x	2	5	6	9
$f(x)$	4	10	12	18
$g(x)$	8	20	24	36

6 Consider the following table of values:
Find:

a $(f + g)(2)$

b $(f + g)(7)$

c $(f \cdot g)(2)$

d $(g - f)(9)$

x	2	7	8	9
$f(x)$	4	14	16	18
$g(x)$	8	28	32	36

7 Given that $f(x) = x - 7$ and $g(x) = 8x + 7$, find the following:

a $(f + g)(x)$

b $(f - g)(x)$

c $(f \cdot g)(x)$

d $\left(\frac{f}{g}\right)(x)$

8 Consider the functions f and g defined by $f(x) = 2x - 2$ and $g(x) = x^2$.

a Find the domain of f.

b Find the domain of g.

c Find the domain of $f + g$.

9 Consider the functions f and g defined by [redacted] $= 2x - 4$ and $g(x) = 3x^2$.

a Find the domain of f.

b Find the domain of g.

c Find the domain of $\dfrac{f}{g}$.

10 Consider $f(x) = x - 3$ and $g(x) = |x|$.

a Find the domain of $(f \cdot g)(x)$.

b Find $(f \cdot g)(x)$.

c Find the domain of $\left(\dfrac{f}{g}\right)(x)$.

d Find $\left(\dfrac{f}{g}\right)(x)$.

e Find the domain of $\left(\dfrac{g}{f}\right)(x)$.

f Find $\left(\dfrac{g}{f}\right)(x)$.

11 Consider $f(x) = \dfrac{9}{x - 7}$ and $g(x) = \sqrt{x - 2}$.

a Find the domain of $f(x)$.

b Find the domain of $g(x)$.

c Find the domain of $(f \cdot f)(x)$.

d Find the domain of $\left(\dfrac{f}{g}\right)(x)$.

e Find $(f \cdot f)(x)$.

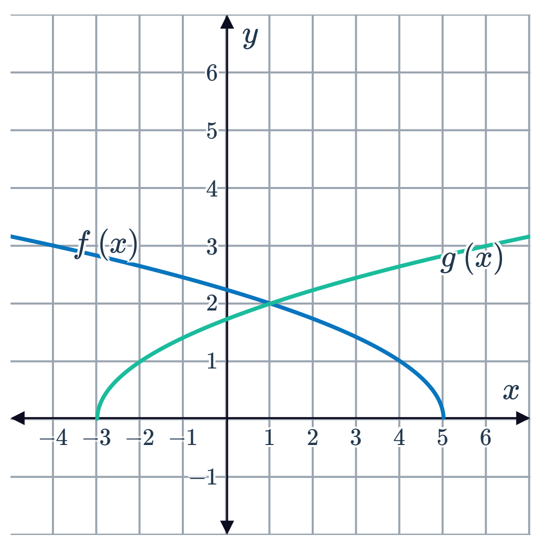
f Find $\left(\dfrac{f}{g}\right)(x)$.

12 The graphs of $f(x) = \sqrt{5 - x}$ and $g(x) = \sqrt{x + 3}$ are shown.
Find the domain of the following functions using interval notation:

a f

b g

c $f + g$



Applications

13 The financial team at The Gamgee Cooperative want to calculate the profit, in dollars, $P(x)$, generated by producing x units of wetsuits. The revenue, in dollars, produced by the product is given by the equation is $R(x) = -\frac{x^2}{4} + 40x$. The cost of production, in dollars, is given by the equation $C(x) = 5x + 410$.

a The profit is calculated as $P(x) = R(x) - C(x)$. Find an equation for $P(x)$ in terms of x.

b Find the values of the following:

i $R(70)$ ii $C(70)$ iii $P(70)$

c Sketch the graphs of $y = R(x)$, $y = C(x)$ and $y = P(x)$ on the same coordinate plane.

14 Elizabeth wants to calculate the cost to travel to Tehran and then Mexico City at certain times of the year. A program on her computer calculates $T(x)$, the total cost of this trip, by adding the cost of travelling to Tehran, $S(x)$, and the cost of traveling to Mexico City, $U(x)$. Based on historical data, the computer program uses the functions:

$$S(x) = x^2 - 200x + 10\,227 \quad \text{and} \quad U(x) = 18x + 15\,423$$

where x is the number of days until the date of travel.

a Find an equation for $T(x)$.

b Find the values of the following:

i $S(91)$ ii $U(91)$ iii $T(91)$

c Sketch the graphs of $y = S(x)$, $y = U(x)$ and $y = T(x)$ on the same coordinate plane.